arpa·e OPEN 2021

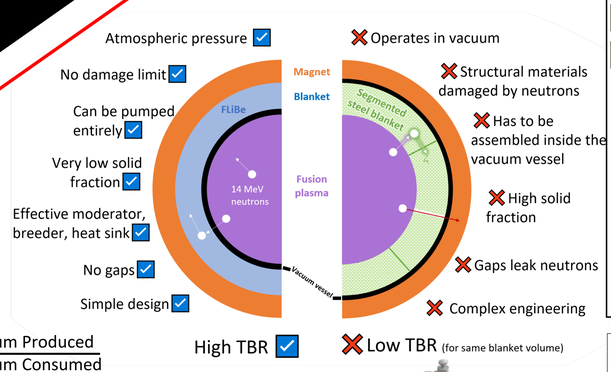
LIBRA

Liquid Immersion Blanket - Robust (tritium) Accountancy

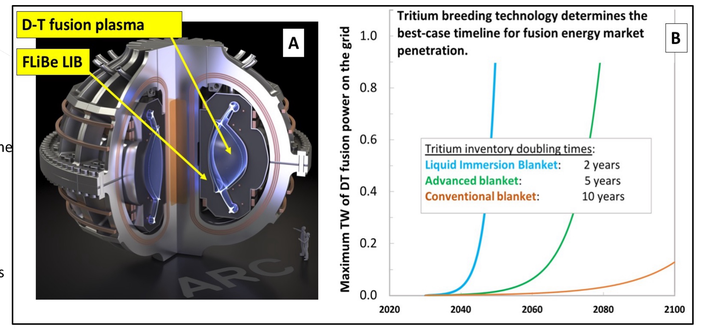
Rémi Delaporte-Mathurin, Collin Dunn, Emily Edwards, Sara Ferry, Nikola Goleš, Edward Lamere, Andrew Lanzrath, Rick Leccacorvi, Samuele Meschini, Ethan Peterson, Stefano Segantin, Rui Vieira, Dennis Whyte, Lihua Zhou, Weiyue Zhou **PI: Kevin Woller**

MIT Plasma Science and Fusion Center

A Liquid Immersion Blanket provides major engineering and tritium breeding advantages to a fusion power plant.



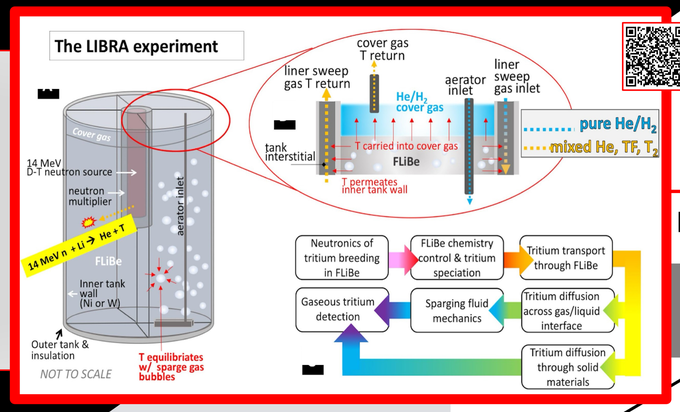
Tritium Breeding Ratio = Tritium Produced / Tritium Consumed



(A) The ARC-class fusion power plant (rendering courtesy of Commonwealth Fusion Systems)
(B) Efficient tritium breeding is fusion's key to rapid energy market penetration

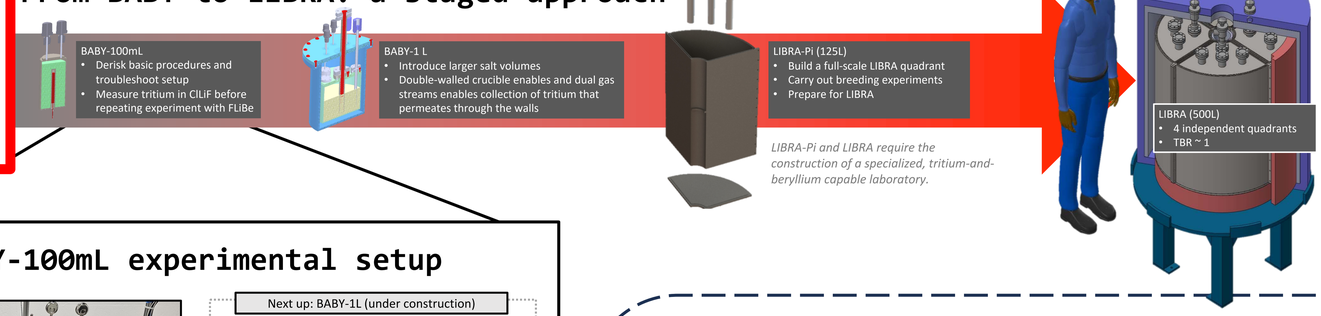
The LIBRA facility

LIBRA will consists of a 14 MeV D-T fusion neutron generator surrounded by **molten FLiBe salt**. It enables the study of tritium breeding and tritium kinetics in an environment representative of a FLiBe liquid immersion blanket for an **ARC-class fusion power plant**. The 4 quadrant tanks operate independently, allowing for repeatability and parametric studies. The planned size of LIBRA is designed to achieve a **global tritium breeding ratio (TBR) of 1** – a world first for a fusion blanket experiment.



2022 LIBRA paper in Fusion Science & Technology

From BABY to LIBRA: a staged approach



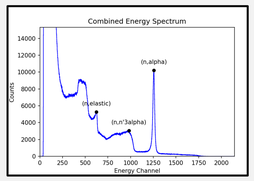
FLiBe tritium breeding experiments have **serious inherent hazards**, including beryllium, hydrofluoric acid creation, large volumes of very high-temperature liquid, 14 MeV neutron radiation fields, and tritium. This incurs lots of expense and requires careful safety planning. This staged approach allows a careful derisking strategy that minimizes exposure to hazards and maximizes the probability of experimental success.



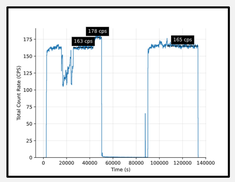
Student Spotlight: Emily Edwards

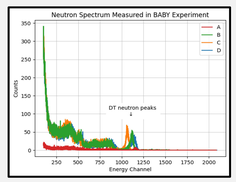
Prior to joining MIT PSFC for her PhD, Emily studied at Abilene Christian University in Texas, where she carried out research at the NEXT lab, which is building a FLiBe-cooled molten salt research reactor.

LIBRA RESEARCH FOCUS
- Neutronics of the BABY/LIBRA experiments
- Characterization of diamond detectors used to monitor energy and fluence of the neutrons emitted by the neutron generators
- Comparison of GEANT4 simulations with experimental results



Neutron spectrum and neutron count rate measured by Cividec Diamond D-T Fusion Neutron Monitor during a BABY experimental run





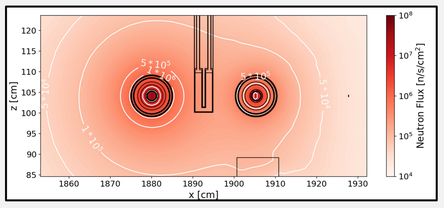
Spectrum measured by proton recoil telescope during a BABY experimental run



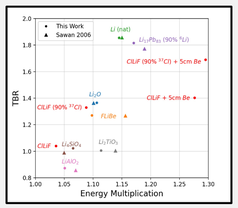
Student Spotlight: Collin Dunn

Prior to joining MIT PSFC for his PhD, Collin studied nuclear engineering at Georgia Tech. He has held internships at Helion, PPPL, and DIII-D.

LIBRA RESEARCH FOCUS
- Neutronics of the BABY/LIBRA experiments
- Quantifying uncertainties experiment simulations due to nuclear data uncertainties
- Exploring novel tritium breeding salts like ClLiF (LiCl + LiF)
- Neutron detection and temperature control systems for BABY/LIBRA

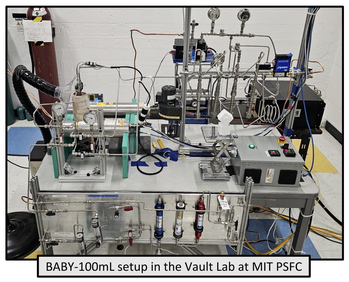


Modeling the expected neutron flux in the BABY experimental setup



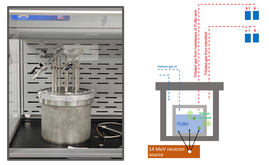
Exploring the potential of alternative salts for tritium breeding in a liquid immersion blanket

BABY-100mL experimental setup

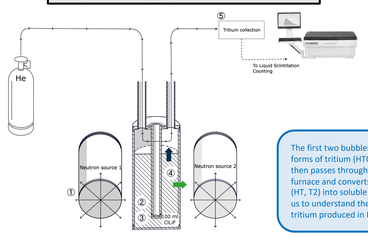


BABY-100mL setup in the Vault Lab at MIT PSFC

Next up: BABY-1L (under construction)

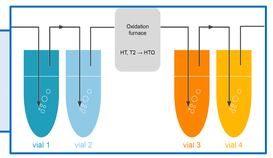


Now that the techniques for neutron detection and tritium collection are refined, we can introduce larger salt volumes and a second gas stream to test for tritium that permeates the crucible walls.



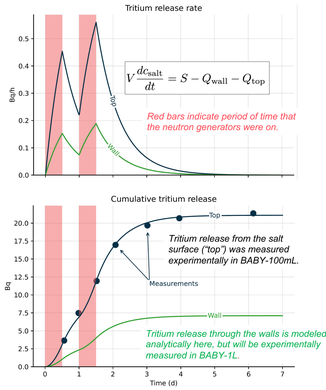
0. Condition with H_2/He and pure He; salt brought to 700C
1. 14 MeV neutrons generated
2. Tritium produced in salt
3. Tritium diffuses/advects through salt
4. Tritium reaches surface and is picked up by sweep gas
5. **Tritium is collected**

The first two bubblers collect all soluble forms of tritium (HTO, TCl/TF). The gas then passes through an oxidation furnace and converts insoluble forms (HT, T2) into soluble forms. This enables us to understand the speciation of the tritium produced in BABY.

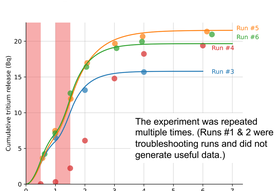


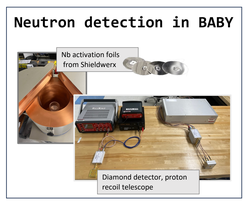
$^6Li + n \rightarrow He + T + 4.8$ MeV
$^7Li + n \rightarrow He + T + n' - 2.5$ MeV

BABY-100mL experimental results



The bubbler vials are regularly changed until no additional tritium is detected. Tritium release rate is modeled analytically and fit to experimental data.





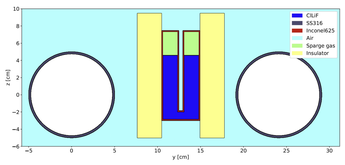
TBR = **3.57E-4** = $\frac{1.17 \times 10^{10} \text{ tritons}}{3.35 \times 10^{13} \text{ neutrons}}$

This is a **true global TBR** measurement, in which all* tritium produced are measured, and all neutrons produced are measured. (It's so small because only 2% of the neutrons impinge on the salt in BABY-100mL).

*BABY-100mL doesn't measure tritium that permeates through the crucible walls, introducing error into this TBR calculation – see simulation results box.

2024 draft of submitted journal paper detailing first results on BABY

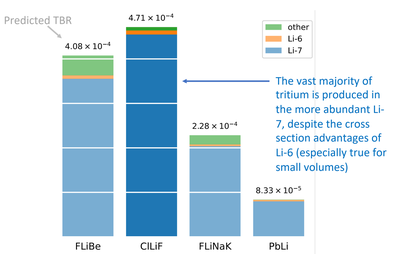
BABY-100mL OpenMC simulation results



Model geometry of the BABY-100mL experimental setup used in OpenMC simulations

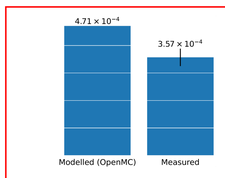
OpenMC

Use of OpenMC for fusion spectrum applications is validated against other neutronics codes in 2024 Fusion Science & Technology paper



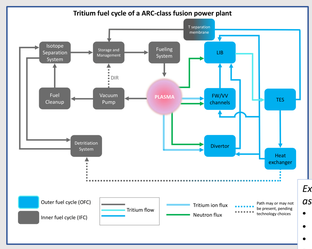
The vast majority of tritium is produced in the more abundant Li-7, despite the cross section advantages of Li-6 (especially true for small volumes)

Predicting TBR for different liquid breeder options in the BABY-100mL set up: ClLiF is the "best" breeder at this volume, but at higher volume, the neutron-multiplying properties of **beryllium** give FLiBe the advantage.



A **discrepancy of ≈25%** was observed between the simulated TBR and the measured TBR. We hypothesize that most of this discrepancy is due to tritium permeating the crucible walls and either remaining there or being released to the lab. **BABY-1L**, **LIBRA-Pi**, and **LIBRA** will use a secondary interstitial sweep gas stream to capture permeated tritium.

LIBRA data → improve accuracy of fuel cycle models

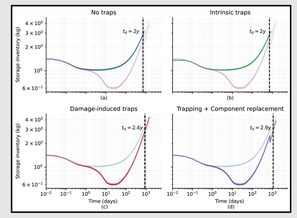


Fuel cycle modeling lets us predict fusion power plant behavior and lets us calculate key design constraints, like the minimum TBR required for the plant to achieve **tritium self-sufficiency**. Experiments like LIBRA refine our understanding of tritium behavior and extractability from breeder materials, leading to improved tools for fusion power plant design.

Example result for an ARC-class plant (no trapping), assuming the following parameters:
- *70% availability factor*
- *Tritium Burn Efficiency of 2%*
- *DIR fraction of 0.3*
- *Overall tritium processing time of 4 hours*
- *Target tritium inventory doubling time of 2 years*

Required TBR: ***1.067***
Required startup inventory: ***1.14 kg***

2023 paper in Nuclear Fusion on modeling the fuel cycles of ARC- and STEP-class fusion power plants



Upcoming work from S. Meschini and team shows that phenomena like tritium trapping in materials has a significant impact on the fuel cycle of fusion power plants. Trapping was not accounted for in the initial modeling effort. This is one example of how fuel cycle modeling is easily rendered inaccurate by lack of data and/or understanding of relevant multiphysics phenomena.

LIBRA is aligned with community research priorities.

Fusion Fuel Cycles/Blankets Research Objectives: Results from the 2023 Fusion Fuel Cycles/Blankets Workshop

Fusion Fuel Cycles Research Objectives:
1. Fueling and Exhaust Processing
2. Isotope processing/rebalancing/storage
3. Tritium confinement/processing/accountancy

Fusion Blankets Research Objectives:
1. Tritium control
2. Functional materials
3. Structural materials
4. Blanket enabling technologies
5. Maintenance and integration

MIT PSFC

June 2024